NOVOGENIA · ACADEMY

FREQUENTLY ASKED QUESTIONS ATHLETIC PERFORMANCE

Answers to typical client questions about the genetic performance analysis.

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01 · TALENT & GENETICS IN SPORTS

How large is the influence of genes on athletic performance?

Estimates vary between 30 and 70%, depending on the discipline. Talent (sprint fibers, VO2max potential, connective tissue stability) is predominantly genetically determined; the training response itself — how strongly the body reacts to a stimulus — also varies genetically by a factor of 2 to 3.

What is the difference between muscle volume and muscle strength?

Volume means more mass (bodybuilder goal) — training with 6-12 reps. Strength means more performance with less mass — training with 1-6 reps at heavy loads. Both training goals require different strategies and different rest periods.

Can I become a successful athlete with "bad" genes?

Yes. Genetics determines talent, but training, consistency and the right strategy make the decisive difference. Knowing your genetic weak points lets you compensate (e.g. antioxidant-rich nutrition, longer rest, careful warm-up).

02 · INJURY RISK & REGENERATION

My report shows unfavourable connective-tissue genes (COL5A1, MMP3). What should I do?

Train with extra care: thorough warm-up, less explosive loading until adapted, longer recovery between intense sessions, and good protein/Vit-C intake to support collagen synthesis. Avoid sudden volume jumps.

Why does the report recommend antioxidants for me?

Intense exercise generates large amounts of free radicals. If your antioxidant genes (SOD2, CAT, GPX) are weak, recovery is slower and micro-inflammation accumulates. Vitamin C, E, polyphenols and selenium directly support these defense systems.

What is a "deload" and how often should I do it?

A deload is a week of gentler training (e.g. first session at half the weight, second at two-thirds). It gives the body recovery time for repair and psychological reset. Typically every ~5 weeks, depending on your genetic regeneration and stress predisposition.

03 · NUTRITION FOR ATHLETES

How does the sport report relate to the nutrition report?

They complement each other. The sport report shows your performance profile, injury risk, and regeneration. The nutrition report tells you which fuel mix (fat/carb/protein) and which supplements support that profile.

Should I take protein powder?

Not necessarily. If you cover 1.6-2.0 g protein per kg body weight through real food and recover well between sessions, no powder is needed. If you struggle to hit those numbers, a clean protein powder can simplify intake.